Name: [redacted] Ortiz

Date: 22 April 2024

Professor: Samuel Chukwuemeka.

Project: Power bill: Residential Electric Service Rate.

Company: Southern Rivers Energy

Website:
https://www.southernriversenergy.com/sites/default/files/pdfs/Residential%20Rate%20Schedule_Revised%202023.pdf

Objectives:

(1.) I will calculate the water bill of the residents of applicable residents within each range of specific electricity usage manually using Arithmetic method.

(2.) I will write the piecewise function of the residential rates for the months of June-September.

(3.) I will recalculate the same electricity bill of the applicable residents for which the company provides electricity within each range of specific electrical energy usage algebraically using the Piecewise Function method.

Information: Customer Charge $30.00 per month
Billing Months: June- September

First 800 kWh or less 13.58¢ per kWh

Over 800 kWh 15.58¢ per kWh

Conversions:

$13.58\ cents\ per\ KWh\ = \frac{13.58}{100} = \$0.01358\ per\ kWh$

$15.58\ cents\ per\ KWh\ = \frac{15.58}{100} = \$0.01558\ per\ kWh$

Arithmetic Method:

Amounts to test are 0 KWh, 500 KWh, 900 KWh, 1500KWh,

(1.) June: 0 KWh

$cost = 30 + 0.01358(500)$

$cost = \$0.01358(0)$

$cost = \$30.00$

(2.) July: 500 KWh

$cost = \$0.01358(500)$

$cost = \$30 + 6.79$

$cost = \$36.79$

(3.) August: 900 KWh

$cost = 30 + 0.01358(800) + 0.01558(900 - 800)$

$cost = 30 + 10.864 + 1.558$

$cost = \$42.422$

$cost \cong \$42.42$

(4.) September: 1500 KWh

$cost = 30 + 0.01358(800) + 0.01558(1500 - 800)$

$cost = 30 + 10.864 + 10.906$

$cost = \$51.77$

Piecewise Function:

Define the variables that we shall use.

Let p be the quantity of power consumed in KWh.

Let c be the cost per unit of power consumed in dollars.

$minimum\ cost = \$30.00$

First Piece: $0 \leq p \leq 800\ KWh;\ \$\ 0.01358\ per\ \ KWh$

$c(p) = 0.01358p + 30$

Second Piece: $p > 800\ KWh;\ \$\ 0.01558\ per\ \ KWh$

$c(p) = 0.01558 + 30$

$c(p) = 30 + 0.01358(800) + 0.01558(p - 800)$

$c(p) = 10.864 + 0.01558p + 30 - 12.464$

$c(p) = 0.01558p + 10.864 + 30 - 12.464$

$c(p) = 0.01558p + 28.4$

$$c(p) = \begin{cases} 0.01358p + 30\,; & 0 \leq p \leq 800 \\ 0.01558p + 28.4\,; & p > 800 \end{cases}$$

Piecewise Function Method:

Amounts to test are 0 KWh, 500 KWh, 900 KWh, 1500KWh.

(1.) June: 0 KWh

$c(p) = 0.01358p + 30$

$c(0) = 0.01358(0) + 30$

$c(0) = \$30.00$

(2.) July: 500 KWh

$c(p) = 0.01358p + 30$

$c(500) = 0.01358(500) + 30$

$c(500) = \$6.79 + \$30$

$c(500) = \$36.79$

(3.) August: 900 KWh

$c(p) = 0.01558p + 28.4$

$c(900) = \$0.01558(900) + 28.4$

$c(900) = \$14.022 + \$28.4$

$c(900) = \$42.422$

$c(900) \cong \$42.42$

(4.) September: 1500 KWh

$(p) = 0.01558p + 28.4$

$c(1500) = \$0.01558(1500) + 28.4$

$c(1500) = \$23.37 + 28.4$

$c(1500) = \$51.77$

Citation: MLA

"Residential Electric Service Rate SCHEDULE: R-8 Farm and Home." *Southernriversenergy*, 1 November 2023, [Schedule R-8 Farm and Home (southernriversenergy.com)]